

7(a)	any seven from: 1 reduced, NAD / FAD ; 2 releases hydrogen / hydrogen splits into proton and electron ; 3 ref. to, inner mitochondrial membrane / cristae ; 4 electrons pass through, electron transport chain / ETC ; 5 ref. to energy release ; 6 protons transferred, through inner membrane / into intermembrane space ; 7 proton gradient established / high proton concentration in intermembrane space ; 8 protons diffuse through ATP synth(et)ase ; 9 ATP produced from ADP and Pi ; 10 ref. to chemiosmosis ; 11 oxygen acts as final electron acceptor (to form water) / described ;	7
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Question	Answer	Marks
7(b)	any four from: 1 population B produces more ATP (than population A) ; ora 2 production of ATP increases at a higher rate for population B (than for population A) ; ora 3 paired data quote ; 4 population B carries out, glycolysis / substrate-linked phosphorylation, and oxidative phosphorylation ; 5 population A carries out (only) glycolysis / respiration in anaerobic conditions / (ethanol) fermentation / substrate-linked phosphorylation or population A cannot carry out oxidative phosphorylation ;	4

Question	Answer	Marks
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